

Week 4: Geometry

Instructions: This handout is meant to go over some basic topics in geometry. The problems covered here are meant to be of the same level as the NMTC descriptive questions and above that. Problems marked at the end with (*) are a bit more tough (and fun!).

The problems covered here are meant to build your problem-solving skills, and as such, they are open-book. You are encouraged to use any textbook, video, or other online or offline resource as long as it doesn't give away the complete solution.

For each section, there are some worked examples. These examples are meant to show the applications of various techniques and how to approach more difficult problems.

It is recommended that you try to attempt all the example problems before reading the solution. Seriously!, since this is a geometry handout, most of the problems rely on angle chasing done on diagrams and then written down on paper. Thus, if you don't have your own diagram to follow along, reading the written down angle chasing becomes really hard.

1. The Basics

This section will cover some basic techniques and intuition that you'll need to tackle introductory geometry problems. It is only assumed that the reader has a strong grip of their school's geometry curriculum (An O level/Matric level knowledge suffices), including various angle, and triangle properties, and the pythagorean theorem.

Consider the following example problem:

Example 1.1 (2011 AMC 10B Problem 18) Rectangle $ABCD$ has $AB = 6$ and $BC = 3$. Point M is chosen on side AB so that $\angle AMD = \angle CMD$. What is the degree measure of $\angle AMD$?

Solution. We start by doing some angle chasing. Letting $\angle AMD = x$, we find that $\angle AMD = \angle MDC$ by alternate angles. We also know that $\angle DMC = \angle AMD = \angle MDC$. Hence $\triangle DMC$ is isosceles, so $MC = DC = 6$.

In $\triangle BCM$, we now have $MC = 6$, $BC = 3$, and $\angle MBC = 90^\circ$. It follows that $\angle BMC = 30^\circ$ (Why?). Hence we have:

$$2x + 30^\circ = 180^\circ \iff x = 75^\circ$$

and we're done!

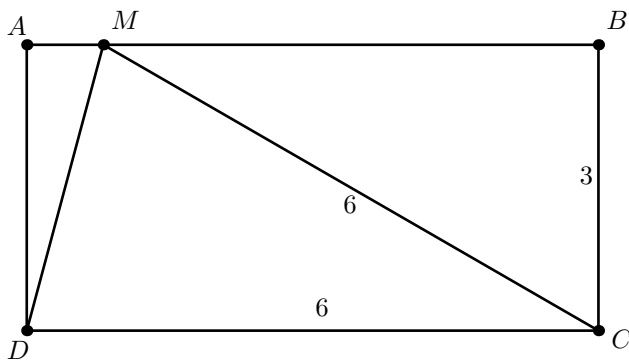


Figure 1.1

Example 1.2 (Angle Bisector Theorem) In $\triangle ABC$. Let D be a point on segment BC . Prove that AD is the angle bisector of $\angle BAC$ if, and only if, $\frac{AB}{AC} = \frac{BD}{CD}$

Solution. Recall that AD being the angle bisector of $\angle BAC$ means that $\angle BAD = \angle DAC$. If you know a bit of trigonometry, you should recognize that we can use the sine rule to work with these equal angles. Using it on $\triangle ABD$ and $\triangle ACD$, We get the following relations

$$\frac{\sin(\angle BAD)}{BD} = \frac{\sin(\angle ADB)}{AB} \quad \text{and} \quad \frac{\sin(\angle DAC)}{CD} = \frac{\sin(\angle ADC)}{AC}$$

But here, $\sin(\angle BAD) = \sin(\angle DAC)$ and $\sin(\angle ADB) = \sin(\angle ADC)$. Hence, $\frac{\sin(\angle BAD)}{\sin(\angle DAC)} = \frac{\sin(\angle ADB)}{\sin(\angle ADC)}$. Now, we re-arrange the above relation to get:

$$\frac{BD}{AB} = \frac{\sin(\angle BAD)}{\sin(\angle ADC)} = \frac{\sin(\angle DAC)}{\sin(\angle ADC)} = \frac{CD}{AC}$$

Which re-arranges to give

$$\frac{BD}{CD} = \frac{AB}{AC}$$

As needed. The proof of the converse is left as an exercise to the reader.

Alternate Solution. Refer to the diagram on the next page. Here's a completely synthetic and way more elegant way to do this. Since the only way we know to get ratios of lengths is through similar triangle, we should try to construct some!

If we let E be the intersection of AD and the line through C parallel AB , we get that $\angle CEA = \angle EAB = \angle DAB$. Now we get that $\triangle BAD \sim \triangle CDE$. Now, if AD is an angle bisector, then

$$\angle EAC = \angle DAC = \angle DAB = \angle EAB = \angle AEC$$

Hence, $\triangle ACE$ is isosceles and we have $CA = CE$.

Finally, $\triangle BAD \sim \triangle CDE$ gives us the ratio

$$\frac{BD}{AB} = \frac{DC}{CE} = \frac{DC}{AC} \iff \frac{AB}{AC} = \frac{BD}{CD}$$

as needed. The proof of the converse is left as an exercise to the reader.

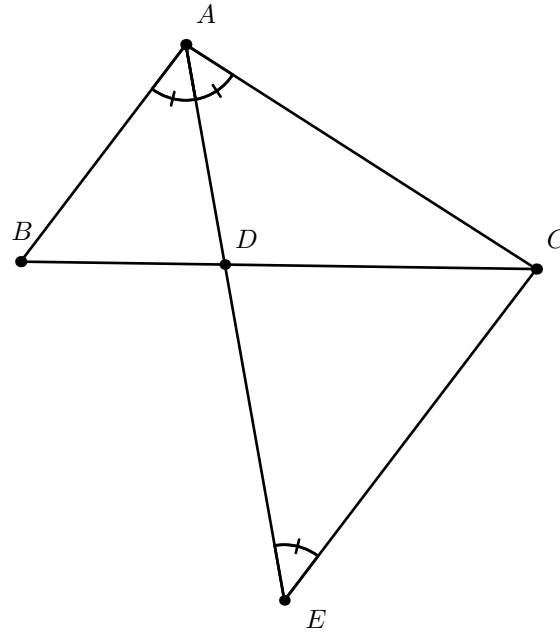


Figure 1.2

Example 1.3 (2022 Pakistan NMTC) Given BE and CF are the altitudes of $\triangle ABC$. P, Q are on BE and the extension of CF respectively such that $BP = AC$, $CQ = AB$. Prove that $AP \perp AQ$.

Solution. Refer to the diagram on the next page. We have to find a way to handle the length conditions on P and Q . We have $CQ = AB$ and $BP = AC$, if we just had $\angle ABP = \angle ACQ$ as well, then by the *SAS* Congruency criteria, we'll get $\triangle ABP \cong \triangle ACQ$. This is equivalent to showing $\angle FBE = \angle FCE$ but this is true as both angles equal $90^\circ - \angle BAC$. Hence, $\triangle ABP \cong \triangle ACQ$. This gives us $\angle BAP = \angle CQA$. Some angle chasing now gives us (You should make your own diagram, and label the angles on your own to follow along):

$$\angle QAP = \angle QAF + \angle FAP = (90^\circ - \angle FQA) + \angle BAP = 90^\circ - \angle CQA + \angle BAP$$

Which is equal to 90° , and we're done.

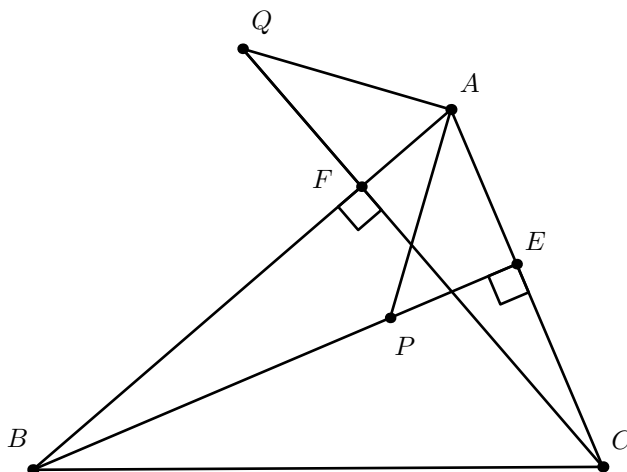


Figure 1.3

Example 1.4 (AIME 2005) In quadrilateral $ABCD$, $BC = 8$, $CD = 12$, $AD = 10$, and $\angle A = \angle B = 60^\circ$. Find AB .

Solution. There are many ways to solve this problem, though most of them require you to construct some new points or lines, below is one such solution with some commentary. Refer to the diagram on the next page.

The first thing to notice is the 60° angles, how do we use these?

A straightforward thing to do would be to try and extend lines AD and BC till they intersect, say at point E . It is now clear that $\triangle AEB$ is equilateral. In order to get a more symmetric (nicer) figure, we can now try drawing the line through C parallel to AB . Let this line intersect AD at F . Notice that $ABCF$ is an isosceles trapezoid. Since $FC \parallel AB$, we have $\triangle FCE$ equilateral. Hence,

$$AB = EB = EC + CB = EC + 8 = FC + 8$$

Now, in $\triangle FDC$, we have that $\angle DFC = 60^\circ$, $FD = 2$, and $DC = 12$. Applying the cosine rule gives us:

$$\begin{aligned} DC^2 &= FD^2 + FC^2 - 2FD \cdot FC \cdot \cos(60^\circ) \\ 140 &= FC^2 - 2FC \\ 141 &= FC^2 - 2FC + 1 \\ &= (FC - 1)^2 \end{aligned}$$

Hence, $FC = \sqrt{141} + 1$ It follows that $AB = \sqrt{141} + 9$

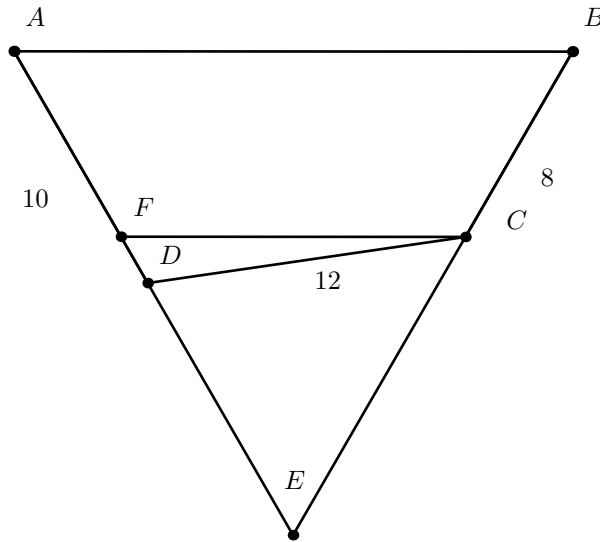


Figure 1.4

The following is probably well known to most of you, but it is still worth mentioning it again:

Theorem 1.1 Let $ABCD$ be a quadrilateral. Then the following statements are equivalent:

- (a) $ABCD$ is a parallelogram.
- (b) $AB \parallel CD$ and $BC \parallel AD$.
- (c) $\angle ABC = \angle CDA$ and $\angle DAB = \angle BCD$.
- (d) The diagonal segments BD and AC bisect each other.

Example 1.5 Let $\triangle ABC$ be a scalene triangle. Point P lies on BC so that AP bisects $\angle BAC$. The line through B perpendicular to AP intersects the line through A parallel to BC at point D . Suppose $BP = 1$ and $PC = 2$. Show that $ABCD$ is a parallelogram.

Solution. Since we have to show $ABCD$ is a parallelogram, it is worth defining the intersection of the diagonals AC and BD as point Y . Further, we let AP and BD intersect at the point X .

Now we know that $\angle AXB = \angle AXY = 90^\circ$ and $\angle BAX = \angle YAX$. Hence $\triangle BAX$ and $\triangle YAX$ are similar. But they both share the side AX , hence they're congruent. So, $AY = AB$.

Now we have to find a way to use the length condition we're given. A straightforward way is through the angle bisector theorem. Indeed, we have

$$\frac{1}{2} = \frac{BP}{PC} = \frac{AB}{AC} = \frac{AB}{AY + YC}$$

Since $AY = AB$, it follows that $YC = AY$ too. So we just need to show that $YB = YD$ and we are done through criteria (d) of Theorem 1.1

But angle chasing gives us $\angle YAD = \angle YCB$. Hence, $\triangle YAD \cong \triangle YCB$. It follows that $YB = YD$ and we're done.

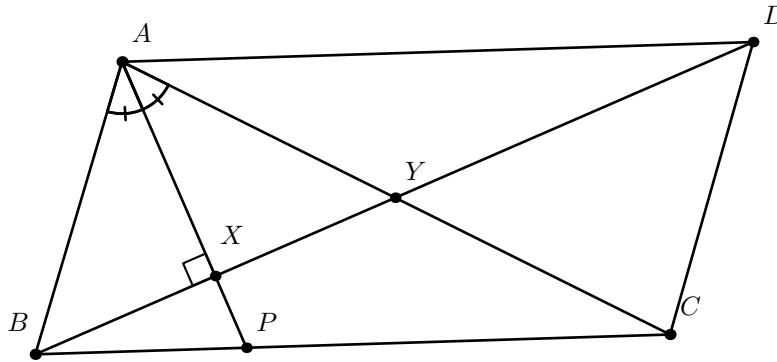


Figure 1.5

Example 1.6 In $\triangle ABC$, P is the midpoint of AB and Q is the midpoint of PC . Let BQ intersect AC at R . Show that $AR = 2RC$.

Solution.

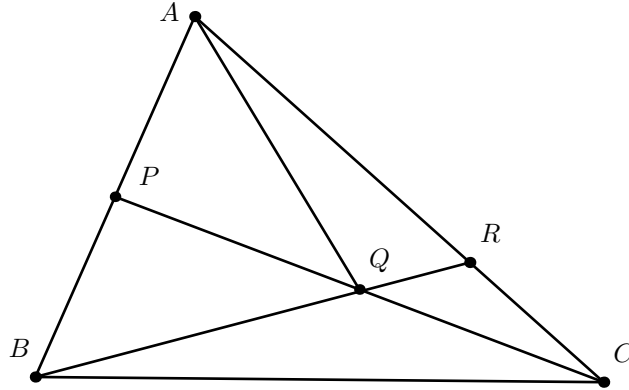


Figure 1.6

We solve this by comparing areas. We denote by $[X]$, the area of X . We'll be using the fact that if two triangles share an altitude, then the ratio of their areas is the ratio of their bases. This fact is quite useful in general. Observe that $\triangle AQP$ and $\triangle BQP$ have the same altitude. Hence,

$$\frac{[AQP]}{[BQP]} = \frac{AP}{BP} = 1$$

Similarly, we have that:

$$\frac{[BPQ]}{[BCQ]} = \frac{PQ}{QC} = 1$$

Hence, $[APQ] = [BQP] = [BCQ]$. We now use the fact that if $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{b} = \frac{c}{d} = \frac{c \pm a}{d \pm b}$ (Verify this!). We now have:

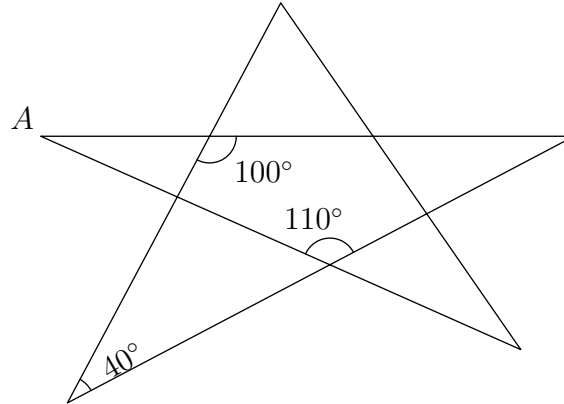
$$\frac{AR}{RC} = \frac{[ABR]}{[CBR]} = \frac{[AQR]}{[CQR]} = \frac{[ABR] - [AQR]}{[CBR] - [CQR]} = \frac{[ABQ]}{[BCQ]} = \frac{[APQ] + [BQP]}{[BCQ]} = 2$$

It follows that $AR = 2RC$ and we're done.

Problems

The problems are ordered roughly in order of difficulty though it may or may not be accurate. You can ask us for hints in the gc or ask for more problems if you aren't satisfied with the ones provided!

Problem 1.1 (1999 AMC 8) Find the degree measure of angle A in the following figure:



Problem 1.2 In isosceles $\triangle ABC$, we have $BC = CA$. Let D be a point such that $BD = BC$ and $BD \perp AC$. If $\angle BAC = 50^\circ$, Find the degree measure of $\angle BDC$.

Problem 1.3 In rectangle $ABCD$, let P be an interior point inside the rectangle. Let DP intersect line AB (extended, if necessary) at the point X and let CP intersect the line DA (extended, if necessary) at the point Y . Suppose that $\angle AXD = \angle DYC$. Show that $DX \perp CY$.

Problem 1.4 (2000 AIME I) In $\triangle ABC$ with $AB = AC$, point D lies strictly between A and C on side $\overline{AC}$, and point E lies strictly between A and B on side $\overline{AB}$ such that $AE = ED = DB = BC$. Find the degree measure of $\angle ABC$.

Problem 1.5 In $\triangle ABC$ with $\angle BAC = 90^\circ$. Let D be a point on AC such that BD is the angle bisector of $\angle ABC$. It is given that $AB = 8$, $BC = 10$. Find DB .

Problem 1.6 (CMIMC 2018) Inside $\triangle ABC$ lies a point P such that $\angle ABP = 20^\circ$ and $\angle ACP = 15^\circ$. Compute the the degree measure of $\angle BPC - \angle BAC$.

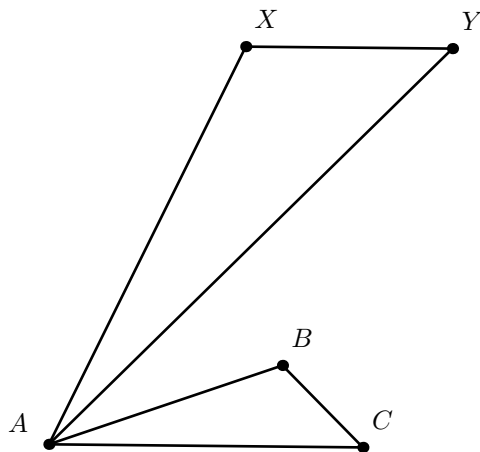
Problem 1.7 (2023 Pakistan NMTC) In $\triangle ABC$, $AB = AC$ and points D , E and F lie on sides AB , BC , and CA respectively such that $DE = EF = FD$. Show that

$$\angle DEB = \frac{1}{2}(\angle ADF + \angle CFE)$$

Problem 1.8 In isosceles $\triangle ABC$, we have $AB = BC$. Let M be the midpoint of segment BC . Let X be a point on AM such that $AX : XM = 2 : 1$. Let BX intersect AC at N . Show that $\triangle BMN$ is isosceles.

Problem 1.9 In $\triangle ABC$, let the angle bisector of $\angle BAC$ intersect side BC at D . Let the line parallel to AB passing through D intersect side AC at E . Show that $\triangle AMD$ is isosceles.

Problem 1.10 (Useful Fact) Suppose $\triangle ABC \sim \triangle AXY$ and both of the triangles have the same orientation as shown. Show that $\triangle ABX \sim \triangle ACY$.



Problem 1.11 In quadrilateral $ABCD$, Let W , X , Y , and Z be the midpoints of sides AB , BC , CD , and DA respectively. Show that $WXYZ$ is a parallelogram.

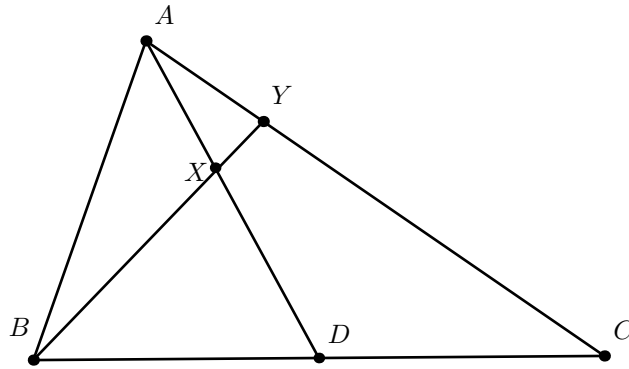
Problem 1.12 In $\triangle ABC$ with $\angle BAC = 90^\circ$. Let D be the foot of the perpendicular from A on side BC . Let E be the foot of the perpendicular from D on side AC . Let M be the midpoint of DC . Given that $AB = 15$ and $AC = 8$, find ME .

Problem 1.13 (Useful Fact) In $\triangle ABC$, Let M , N , and L be the midpoints of sides BC , CA , and AB respectively. The lines AM , BN , and CL are called medians of $\triangle ABC$ and they concur (meet) at a point called the centroid, G .

- (a) Show that the six triangles $\triangle BGM$, $\triangle CGM$, $\triangle CGN$, $\triangle AGN$, $\triangle AGL$, and $\triangle BGL$ have equal area.
- (b) Show that the centroid of a triangle divides a median in a ratio 2 : 1.

A Hint¹

Problem 1.14 In $\triangle ABC$, AD is a median. X is a point on AD such that $AX : XD = 2 : 3$. Let the line BX intersect AC at Y . Show that $YC = 3AY$. Further, show that $BX = 4XY$.



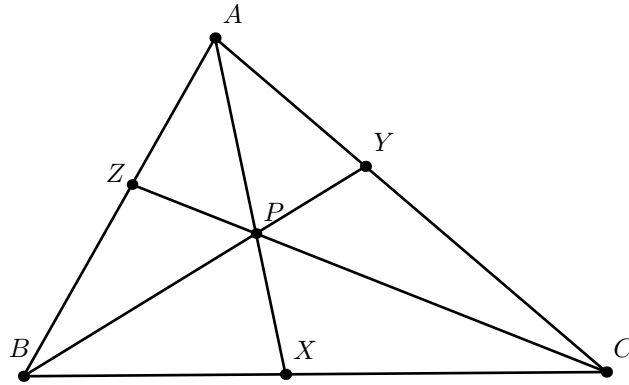
Problem 1.15 Let $ABCD$ be a parallelogram with area 1. Let M be the midpoint of the side AD . Let lines BM and AC intersect at P . Find the area of $MPCD$.

Problem 1.16 (Ceva's Theorem) In a triangle, a cevian is a line joining a vertex of the triangle to a point on the opposite side of the triangle. In $\triangle ABC$, Let lines AX , BY , and CZ be cevians which concur (meet) at a point P . (See the diagram on the next page). Show that we have ²

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = 1$$

¹Try comparing areas like we did earlier.

²In fact, the converse of this is also true. If the given ratio holds, then the cevians concur.



Problem 1.17 In triangle ABC , Let D and E be the midpoints of BC and AC respectively. Suppose $AD \perp BE$. Find the length of AB , if $BC = 6$ and $AC = 8$

Problem 1.18 Let $ABCD$ be a quadrilateral with $BC = 10$, $AD = 15$, and $AB = CD$. If $\angle A = 50^\circ$ and $\angle D = 40^\circ$, find the area of $ABCD$.

Problem 1.19 (2010 AMC 12A Problem 8) $\triangle ABC$ has $AB = 2 \cdot AC$. Let D and E be on $\overline{AB}$ and $\overline{BC}$, respectively, such that $\angle BAE = \angle ACD$. Let F be the intersection of segments AE and CD , and suppose that $\triangle CFE$ is equilateral. Find $\angle ACB$.

Problem 1.20 (2002 AMC 12B Problem 20) Let $\triangle XOY$ be a right-angled triangle with $\angle XOY = 90^\circ$. Let M and N be the midpoints of sides OX and OY , respectively. Given that $XN = 19$ and $YM = 22$, find XY .

Problem 1.21 (2003 AMC 10A Problem 22) In rectangle $ABCD$, we have $AB = 8$, $BC = 9$, H is on BC with $BH = 6$, E is on AD with $DE = 4$, line EC intersects line AH at G , and F is on line AD with $GF \perp AF$. Find the length of GF .

Problem 1.22 (RMO 1993) Let ABC be an acute angled triangle and CD be the altitude through C . If $AB = 8$ and $CD = 6$, find the distance between the midpoints of AD and BC .

Problem 1.23 (BMO1 1995) $\triangle ABC$ has a right angle at C . The angle bisectors of $\angle BAC$ and $\angle ABC$ meet BC and CA at P and Q respectively. The points M and N are the feet of the perpendiculars from P and Q to AB . Find $\angle MCN$.

Problem 1.24 (Incentre Config) In $\triangle ABC$, let I be the intersection of the angle bisectors of $\angle ABC$ and $\angle ACB$.

- (a) Let the perpendicular from I intersect sides BC , CA , and AB at points D , E , and F respectively. Show that $IE = ID = IF$.
- (b) Show that AI is the angle bisector of $\angle BAC$. (This is equivalent to proving that the three angle bisectors of a triangle concur! Indeed, their concurrence point is called the incentre and we'll encounter this point again quite often).
- (c) Show that $\angle BIC = 90 + \frac{1}{2}\angle BAC$

Problem 1.25 (OMM 2009) In $\triangle ABC$, let D be the foot of the altitude from A to BC . A circle centered at D with radius AD intersects lines AB and AC at P and Q , respectively. Show that $\triangle AQP \sim \triangle ABC$.

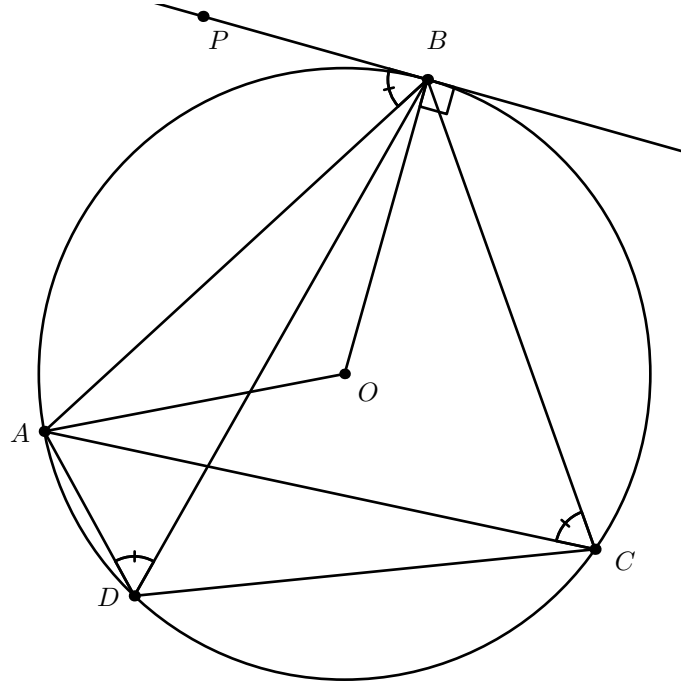
Problem 1.26 (AIME 2007) Square $ABCD$ has side length 13, and points E and F are exterior to the square such that $BE = DF = 5$ and $AE = CF = 12$. Find EF^2 .

Problem 1.27 (AIME 2023) Let $\triangle ABC$ be an isosceles triangle with $\angle A = 90^\circ$. There exists a point P inside $\triangle ABC$ such that $\angle PAB = \angle PBC = \angle PCA$ and $AP = 10$. Find the area of $\triangle ABC$.

Problem 1.28 In $\triangle ABC$, M is the midpoint of AC , the angle bisector of $\angle BAC$ intersects BC at P and BM at N . Show that $AP \perp BM$ if, and only if, $PC = 2PM$.

2. Angle Chasing!

This section is a bit advanced but explores the art of angle chasing. The soul of this section is the following result regarding cyclic quadrilaterals. Refer to the following diagram for the upcoming text.



Theorem 2.1 Let $ABCD$ be a quadrilateral. Then the following statements are equivalent.

- (a) A, B, C , and D are concyclic (lie on a circle).
- (b) $\angle ABC + \angle ADC = 180^\circ$
- (c) $\angle ACB = \angle ADB$

We call $ABCD$ a cyclic quadrilateral and here are more related facts.

Theorem 2.2 (Inscribed Angle Theorem) Let the circle which $\triangle ABC$ is inscribed in have center O . If O and C lie on the same side of the line AB , then $\angle AOB = 2\angle ACB$. Otherwise, we have $\angle AOB = 360 - 2\angle ACB$.³

³Note that this implies that $\angle ACB = 90^\circ$ if, and only if, AB is a diameter of the circle.

Theorem 2.3 (Tangent Criterion) Let $\triangle ABC$ be inscribed in a circle Γ with center O and let $P \neq B$ be any point. Then the following statements are equivalent.

- (a) AP is tangent to Γ .
- (b) $\angle PBA = \angle BCA$
- (c) $OB \perp PB$

We will now see how cyclic quadrilaterals arise everywhere!

Example 2.1 (Orthocentre Config) In $\triangle ABC$, let D , E , and F be the feet of the altitudes from A , B , and C respectively. These three altitudes concur at a point H , called the orthocentre. Show that the six quadrilaterals $BFHD$, $CEHD$, $AEHF$, $BCEF$, $AFDC$, and $AEDB$ are all cyclic.

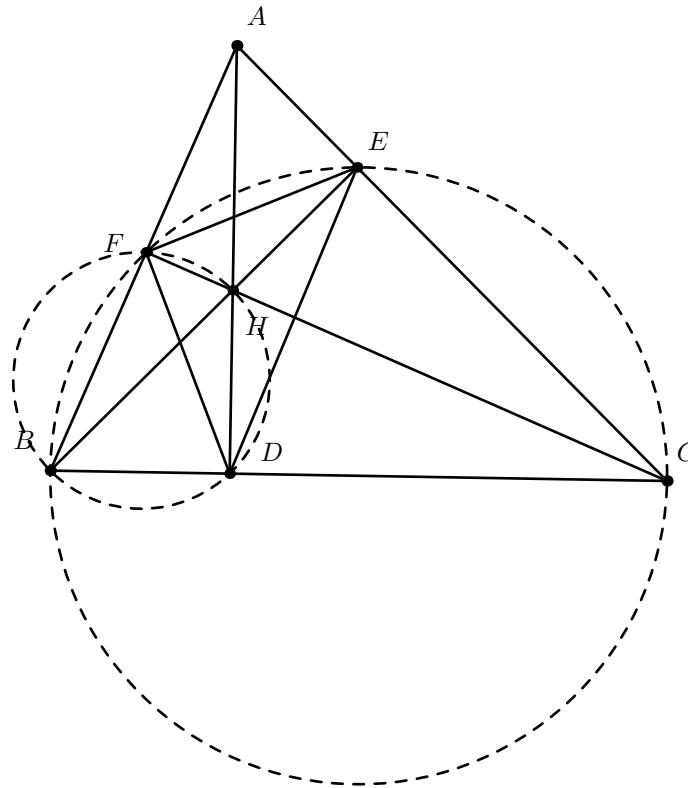


Figure 2.1

Only a few of the cyclic quadrilaterals are shown in the above diagram.

Solution. To show $BFHD$ is cyclic, observe that

$$\angle BFH = \angle BFC = 90^\circ = \angle BDA = \angle BDH$$

Hence, by Theorem 2.1 (c), $BFHD$ is cyclic and similarly, $CEHD$ and $AEHF$ are also cyclic quadrilaterals.

To show $BCEF$ is a cyclic, observe that

$$\angle BFC = \angle BEC = 90^\circ$$

Similarly, $AFDC$ and $AEDB$ are also cyclic quadrilaterals.

We can also show that BC is a diameter of circle $BFEC$ and similarly, BH is a diameter of circle $BFHD$.

Example 2.2 (Incentre-Excentre Lemma) In $\triangle ABC$ with incentre I , and circumcircle (The circle which $\triangle ABC$ is inscribed in) Γ , let the midpoint of the arc BC of Γ not containing A be M .

- (a) Then, M is the centre of the circle passing through B, I, C .
- (b) Let I_A be the reflection of I over M , Then $I_A B$ and $I_A C$ bisect the exterior angles $\angle ABC$ and $\angle ACB$ respectively.

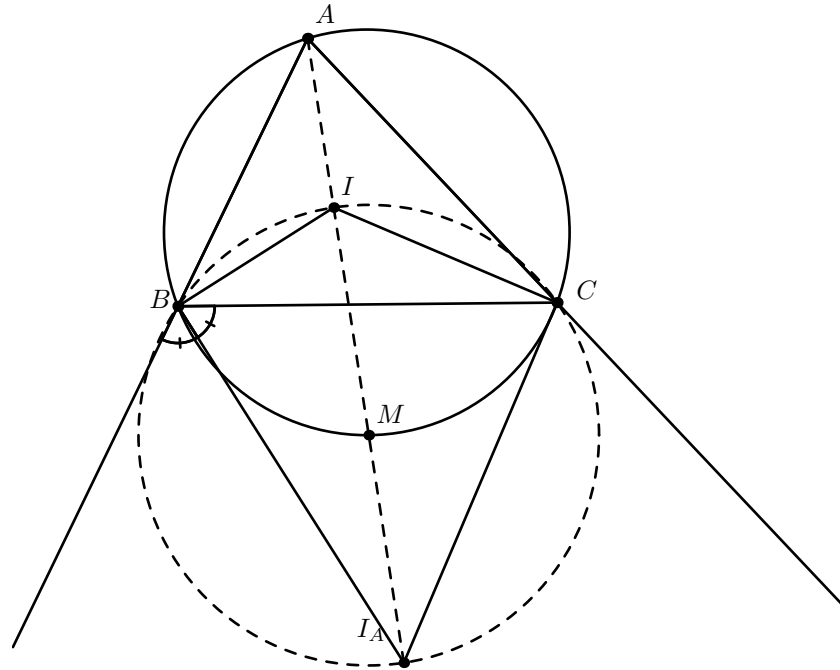


Figure 2.2

Solution. For convenience, let's give the angles of $\triangle ABC$ some names. Let $\angle BAC = 2\alpha$, $\angle ABC = 2\beta$, and $\angle ACB = 2\gamma$. Notice that $\alpha + \beta + \gamma = 90^\circ$.

- (a) We now do some angle chasing using Theorem 2.1, Since M is the midpoint of the arc BC , then $MB = MC$, but $\angle BMC = 180 - \angle BAC = 180 - 2\alpha$. It follows that $\angle MCB = \angle MBC = \alpha$. Now we have:

$$\angle MCB = \angle MAC = \alpha = \angle IAC$$

It follows that A, I, M are collinear (lie on the same line). (why?).
Now, we have to show $MB = MI = MC$, we first show $MB = MI$.
This is equivalent to showing $\angle MBI = \angle MIB$. Note that

$$\angle MBI = \angle MBC + \angle IBC = \alpha + \beta$$

And also:

$$\angle MIB = 180^\circ - \angle BIA = 180^\circ - (90^\circ + \frac{1}{2}\angle BCA) = 90^\circ - \gamma = \alpha + \beta$$

Where $\angle BIA = 90^\circ + \frac{1}{2}\angle BAC$ follows by Problem 1.24(c). Hence, $\angle MBI = \angle MIB$ and $MB = MI$. similarly, we can show $MI = MC$ and we're done.

- (b) We will show $I_A B$ bisects the exterior angle of $\angle ABC$, as the other part is similar.

Since $I_A I$ is a diameter of the circle passing through B, I, C, I_A , it follows that $\angle I_A B C + \angle I B C = 90^\circ$. Hence, $\angle I_A B C = 90 - \beta$. Since the exterior angle of $\angle ABC = 180 - 2\beta$. It is twice $\angle I_A B C$. Hence, $I_A B$ bisects the exterior angle and we're done.

Okay, enough theory now. Let's do some actual contest problems!

Example 2.3 (2013 JBMO G1) Let AB be a diameter of a circle ω and center O , OC a radius of ω perpendicular to AB , M be a point of the segment (OC) . Let N be the second intersection point of line AM with ω and P the intersection point of the tangents of ω at points N and B . Prove that points M, O, P, N are cocyclic.

Solution. This problem consists of straightforward applications of all the Theorems 2.1-2.3.

AB is a diameter, so we have

$$\angle MNB = \angle ANB = 90^\circ = \angle MOB$$

Hence, $MNBO$ is cyclic. Now, for convenience, let $\angle NBA = x$. Next, by Theorem 2.3, we know that

$$\angle PBN = \angle PNB = \angle NAB = 90^\circ - \angle NBA = 90^\circ - x$$

It follows that $\angle NPB = 2x$. But we also have

$$\angle NOB = 2\angle NAB = 2(90^\circ - x) = 180^\circ - 2x$$

. Hence, $\angle NOB + \angle NPB = 180^\circ$. So, $NPBO$ is cyclic. This combined with the fact that $MNBO$ is cyclic gives us the desired result.

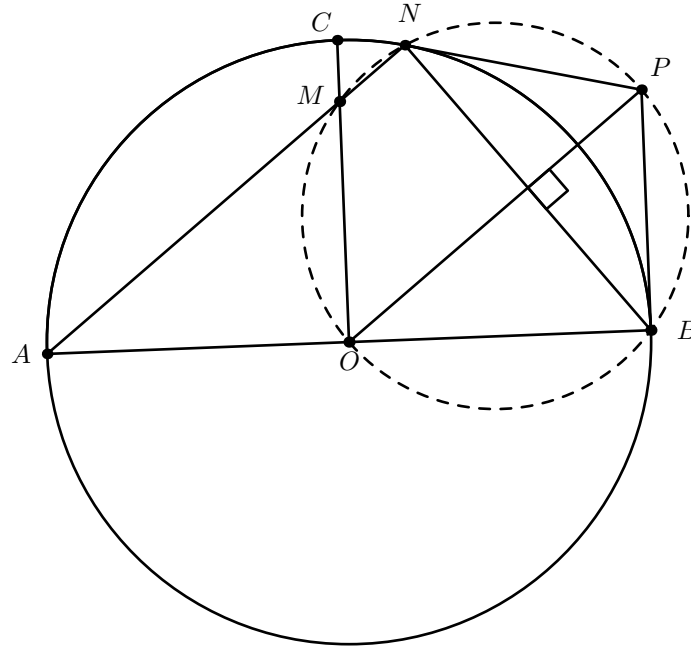


Figure 2.3

Finally, we attempt a problem from the IMO.

Before that, recall the fact that a circumcircle of a triangle is the circle which the triangle is inscribed in. Further, note that the circumcentre (the center of the circumcircle) of a triangle is the intersection of the three perpendicular bisectors of that triangle. Try to prove this if you haven't!

Example 2.4 (2004 IMO P1) Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC . The bisectors of the angles $\angle BAC$ and $\angle MON$ intersect at R . Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC .

Solution. See the next page for a final diagram (Note that the diagram actually spoils parts of the problem!).

For problems like these it is very useful to draw a large and well-drawn diagram, both in order to not get lost in the complexity, but also to guess things regarding the problem. Evan Chen has a magnificent guide on diagram construction linked [here](#).

Let's start by focusing on point R . We have that $OM = ON$ and OR is the angle bisector of $\angle MON$. It follows that OR is the perpendicular bisector of MN . R also lies on the angle bisector of $\angle MAN$. If you're familiar with the Incentre-Excentre Lemma, this should convince you that R is the midpoint of the arc MN in the circumcircle of $\triangle AMN$.

But proving this directly is a bit tricky, we instead define a phantom point R' as the intersection of the circumcircle of $\triangle AMN$ and the angle bisector of $\angle A$. If we can show that R' lies on the perpendicular bisector of MN , we'll have that $R = R'$. This is equivalent to showing that $R'M = R'N$ but this is true as $\angle R'MN = \angle R'AN = \angle R'AM = \angle R'NM$. Hence, $R = R'$. This technique is called the use of phantom points and can be quite useful. We now know that $ANRM$ is cyclic.

Next, if you have a good diagram of the problem at hand, you'll see that the supposed point which is the intersection of the circumcircles of BMR and CNR looks like it lies on the A -angle bisector. Let's try to prove this. Let the intersection of the A -angle bisector and line BC be D . We will individually show that D lies on both of the wanted circles. To show that $BMRD$ is cyclic, we angle chase to get:

$$\begin{aligned}\angle MRD &= 180^\circ - \angle MRA \\ &= 180^\circ - \angle MNA \\ &= \angle MNC \\ &= 180^\circ - \angle MBC \\ &= 180^\circ - \angle MBD\end{aligned}$$

So, $\angle MRD + \angle MBD = 180^\circ$, hence $BMRD$ is cyclic. Similarly, we can show that $CNRD$ is cyclic and we're done.

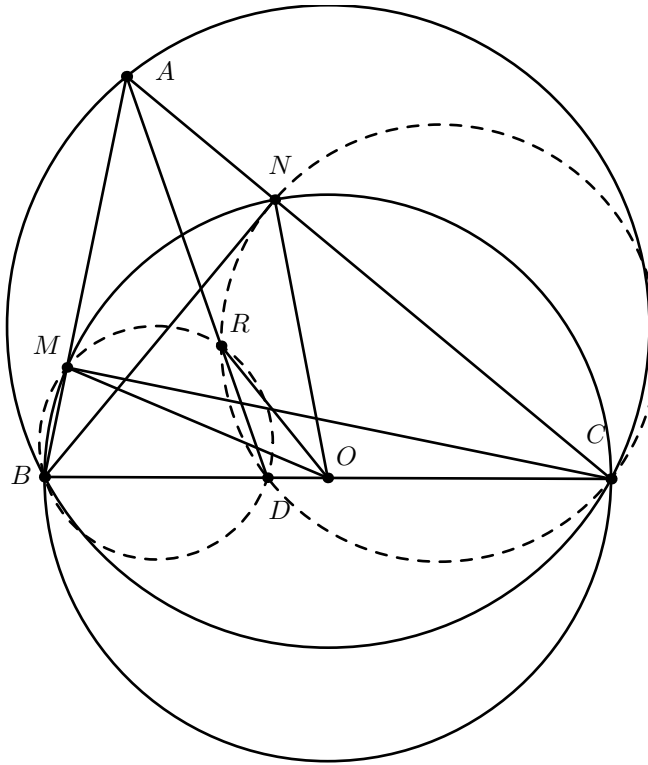


Figure 2.4: IMO 2004/1

Problems

The problems are ordered roughly in order of difficulty though it may or may not be accurate. You can ask us for hints in the gc or ask for more problems if you aren't satisfied with the ones provided!

Problem 2.1 (2024 Pakistan NMTC) In $\triangle ABC$ with circumcircle Γ , let P , Q , and R be points on the arcs BC , CA , and AB respectively of Γ that don't have the opposite vertex of the triangles. Show that

$$\angle ARC + \angle CQB + \angle BPA = 360$$

Problem 2.2 (Shahmeer Khan) In $\triangle ABC$ with orthocentre H , let the circle centered at H passing through A intersect lines AB and AC at P and Q respectively. Show that points P, Q, C and B lie on a circle.

Problem 2.3 In $\triangle ABC$, let P be a point on side BC . The line parallel to C through P intersects the tangent to the circumcircle of $\triangle ABC$ at a point Q . Show that $APBQ$ is cyclic.

Problem 2.4 In a cyclic quadrilateral $ABCD$, the diagonals AC and BD intersect at P . Let the circumcentre of $\triangle BAP$ be O . Show that $OP \perp AC$.

Problem 2.5 In $\triangle ABC$ with circumcircle Γ , suppose we had a line t which is parallel to the tangent to Γ at C . Let t intersect sides AC and CB at X and Y respectively. Show that $BAXY$ is cyclic.

Problem 2.6 In $\triangle ABC$ with circumcircle Γ , suppose that we have $\angle BAC + \angle BCA = 2\angle ABC$. Show that the reflection of the circumcentre of $\triangle ABC$ over BC lies on Γ .

Problem 2.7 (Incicle) In $\triangle ABC$ with incentre I , let the foot of the perpendicular from I to BC be D . Show that the incircle (The circle which is inscribed inside of $\triangle ABC$) has centre I and radius ID .

Problem 2.8 (Orthocentre Config) In $\triangle ABC$ with altitudes D, E, F on sides BC, CA , and AB respectively. Let H be the orthocentre of $\triangle ABC$. Show that H is the incentre of $\triangle DEF$.

Problem 2.9 (2025 Pakistan TSTST P1) Let $ABCD$ be a cyclic quadrilateral. Let the lines CD and BA meet at E . The line through D which is tangent to the circumcircle of ADE meets the line CB at F . Prove that $\triangle CDF$ is isoscles.

Problem 2.10 Let O and H denote the circumcenter and orthocenter of an acute $\triangle ABC$, respectively. Show, that $\angle BAH = \angle CAO$.

Problem 2.11 (Miquel Point) In $\triangle ABC$, points D , E , and F lie on sides BC , CA and AB respectively. Show that there exists a point lying on all three circumcircles of $\triangle AEF$, $\triangle BFF$, and $\triangle CDE$. (Note that there are alot of possible configurations so you should just do one and call it a day. Just trust me that all the configurations work.)

Problem 2.12 In $\triangle ABC$ with orthocentre H , show that the reflection of H over BC and over the midpoint of BC lie on the circumcircle of $\triangle ABC$.

Problem 2.13 Two circles intersect at A and B . One of their common tangents touches the circles at P and Q . Let A' be the reflection of A across the line PQ . Prove that $A'PBQ$ is a cyclic quadrilateral.

Problem 2.14 Let $ABCD$ be a square. Points P and Q , respectively, are chosen on sides BC and CD such that $|\angle QAP| = 45^\circ$. Let $X = AP \cap BD$, $Y = AQ \cap BD$. Show, that X, Y, P, Q, C are concyclic.

Problem 2.15 Circles k, l are given, intersecting each other at A, B . Let K, L , in order, be points of tangency of their common tangent constructed in such a way in which point B is inside $\triangle AKL$. Choose points M, N on k, l such that A is in the interior of $\overline{MN}$. Prove, that $KLMN$ is cyclic if and only if MN is tangent to the circumcircle of $\triangle AKL$.

Problem 2.16 Let ABC be an acute triangle. Let $\overline{BE}$ and $\overline{CF}$ be altitudes of $\triangle ABC$, and denote by M the midpoint of $\overline{BC}$. Prove that $\overline{ME}, \overline{MF}$ and the line through A parallel to $\overline{BC}$ are all tangents to the circumcircle of $\triangle AEF$.

Problem 2.17 Show that a quadrilateral $ABCD$ can have a circle Γ inscribed inside of it (Γ is tangent to all four sides of the quadrilateral) if, and only if $AB + CD = BC + AD$.

Problem 2.18 (2019 ONMAPS P7) Let $ABCD$ be a square. The semi-circle of diameter AB drawn inside the square intersects the circle of center C and radius CB at P . The line CP intersects the circumcircle of triangle APD at Q (not at P). Prove that $\angle BCQ = 2\angle DQC$.

Problem 2.19 (Simson Line) In $\triangle ABC$, let P be any point. Let the feet of the perpendicular from P to lines BC , CA , and AB be X , Y , Z respectively. Show that P lies on the circumcircle of $\triangle ABC$ if, and only if, X , Y , and Z are collinear.

Problem 2.20 (Power of a Point) Let Γ be a circle and let P be any point not on Γ . Let k , l , and m be lines passing through P . Suppose that k intersects Γ at A , and B , l intersects Γ at X and Y , and m is tangent to Γ at the point T .

- (a) Show that $PA \cdot PB = PX \cdot PY = PT^2$. Use this to conclude that the quantity $PA \cdot PB$ is constant for all lines k .
- (b) Let O be the centre of Γ . Show that $PA \cdot PB = |PO^2 - OA^2|$.

Problem 2.21 (Converse of Power of a Point) Let A, B, C, D, P be any points in the plane. Suppose that either P lies on both segments AB and CD or in neither segment. If $PA \cdot PB = PC \cdot PD$, show that A, B, C , and D are concyclic. This result is quite powerful as it allows us to find cyclic quadrilaterals based on length.

Problem 2.22 In $\triangle ABC$, let M be the midpoint of BC . Let Q be the foot of the perpendicular from the orthocentre H to AM .

- (a) Show that $MQEC$ is cyclic.
- (b) Show that BC is tangent to the circumcircle of $\triangle AQC$ and $\triangle AQB$.
- (c) Let A' be the reflection of A over M . Show that $BHQA'$ is cyclic.

The above point Q is called the $A - HM$ point of $\triangle ABC$.

Problem 2.23 In $\triangle ABC$, let D, E and F be the feet of the perpendiculars from A, B and C to BC, CA and AB respectively. Let P, Q, R and S be the feet of the perpendiculars from D to BA, BE, CF and CA respectively. Prove that P, Q, R and S are collinear.

Problem 2.24 In $\triangle ABC$, let M be the midpoint of side AC . The circle tangent to BC at B and passing through M meets the line AB again at $P \neq B$. Show Prove that $AB \cdot BP = 2BM^2$.

Problem 2.25 (2025 Pakistan TST P2) In $\triangle ABC$, circle k_1 passes through points A and B and is tangent to line AC . Similarly, circle k_2 passes through A and C and is tangent to AB . Circle k_1 intersects line BC at point $D \neq B$ and meets circle k_2 at point $E \neq A$. Prove that line DE bisects segment AC .

Problem 2.26 In acute $\triangle ABC$, with circumcenter O , the circumcircle of $\triangle BOC$ and the circle with diameter AC intersect again at point P ($P \neq C$). Point Q is on segment PC such that $PB = PQ$. Prove that $\angle AQP = \angle ABC$.

Problem 2.27 (2024 INMO P1) In triangle ABC with $CA = CB$, point E lies on the circumcircle of ABC such that $\angle ECB = 90^\circ$. The line through E parallel to CB intersects CA in F and AB in G . Prove that the center of the circumcircle of triangle EGB lies on the circumcircle of triangle ECF .

Problem 2.28 (Shahmeer Khan)(*) In $\triangle ABC$, with orthocenter H , let M be the midpoint of BC . Let Γ be the circle centered at H passing through A and let Ω be the circumcircle of $\triangle BHC$. Let AM intersect Γ at P , and AM intersect Ω at Q . The tangent to Ω at Q intersects lines AB and AC at X and Y respectively. Let Z be the reflection of Y over Q . Prove that points X, Z, P , and A lie on a circle. A Hint⁴

Problem 2.29 (2022 EGMO P1) Let ABC be an acute-angled triangle in which $BC < AB$ and $BC < CA$. Let point P lie on segment AB and point Q lie on segment AC such that $P \neq B$, $Q \neq C$ and $BQ = BC = CP$. Let T be the circumcenter of triangle APQ , H the orthocenter of triangle ABC , and S the point of intersection of the lines BQ and CP . Prove that T, H , and S are collinear.

Problem 2.30 (Shahmeer Khan) In $\triangle ABC$ with incentre I , Let the A -angle bisector intersect side BC at D . Let the intersection of line BI and the perpendicular bisector of segment AD be O . Show that the circle centered at O passing through D , and the circumcircle of $\triangle AIC$ have a common point lying on line BC .

⁴Use Problem 2.20-2.21 to find an equivalent statement for the conclusion. Further hint, introduce the points from Problem 2.2

Problem 2.31 (2025 Bulgaria EGMO TST) The point D on side BC of triangle ABC ($BC > AB$) is such that $\angle ADB = \angle BAC$. The angle bisectors of $\angle ACB$ and $\angle ADB$ intersect at L , and AK ($K \in BC$) is the angle bisector $\angle BAC$. The circumcircle of triangle ADK intersects the line AB for the second time at S , and the circumcircle of triangle ASL intersects the line CL for the second time at T . Prove that the line ST bisects the segment AK .

Problem 2.32 (2018 IMO P1)(*) Let Γ be the circumcircle of acute triangle ABC . Points D and E are on segments AB and AC respectively such that $AD = AE$. The perpendicular bisectors of BD and CE intersect minor arcs AB and AC of Γ at points F and G respectively. Prove that lines DE and FG are either parallel or they are the same line.